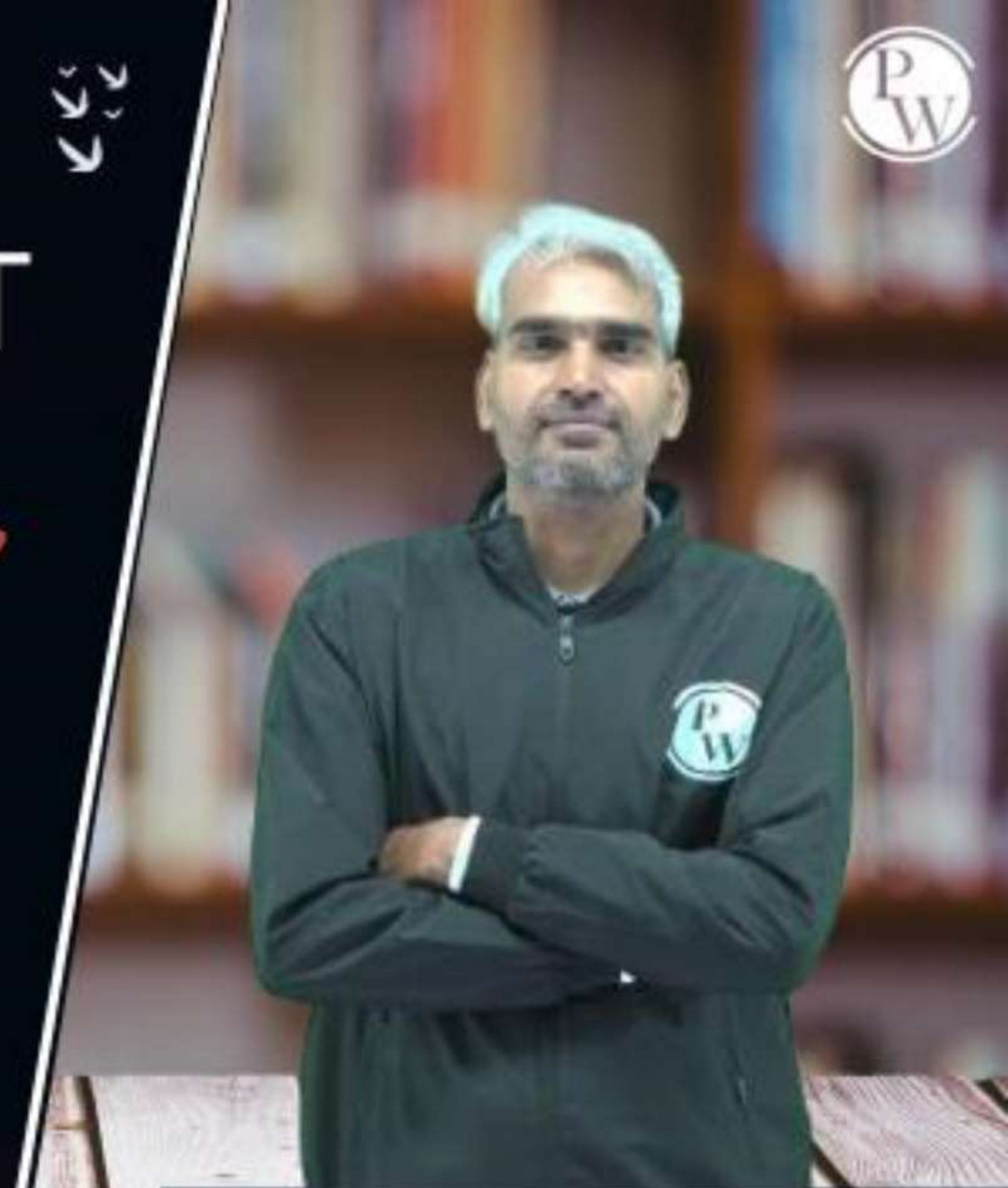
Computer Science & IT

Discrete Mathematics

Graph Theory

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture

- **Topic** Graph isomorphism
- **Topic** Self-complementary graph

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Topics to be Covered

Planar Graphs

Topic	Planar graphs
Topic	Region and degree of regions
Topic	Sum of degree theorem for regions
Topic	Simple planar graph
Topic	Euler's equation

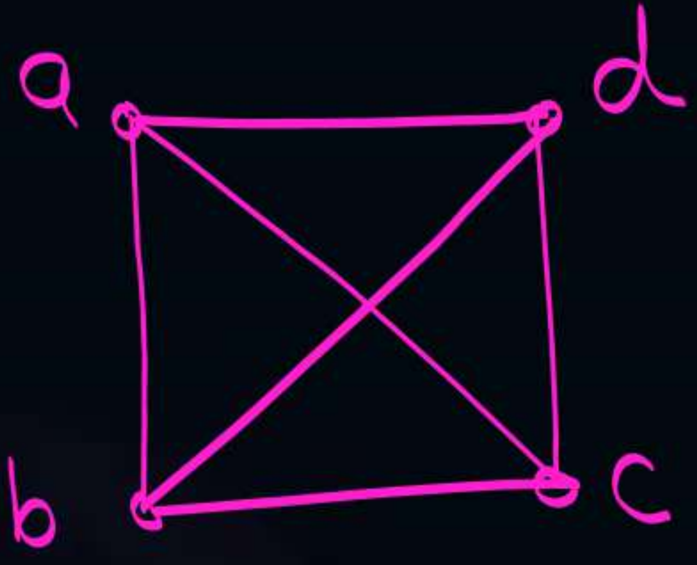
Slide

Topic : Planar graph

* A graph G= (V, E) is said to be planar if it can be drawn in the plane so that no two edges of G intersect each other at a non-vertex point.

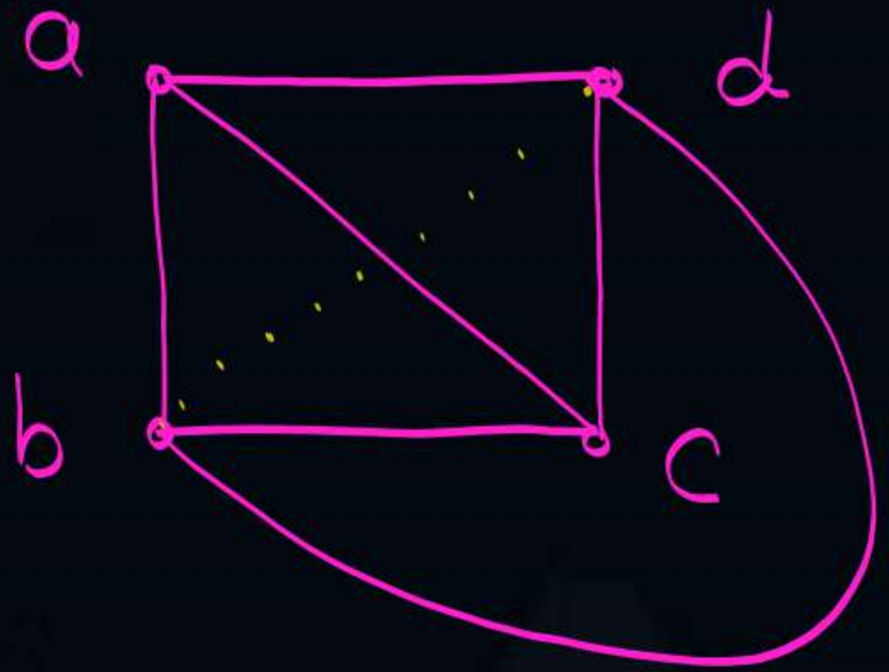
  Such a drawing of a planar graph is called a planar embedding of the graph.

Topic : Planar graph



K_4

$\Longrightarrow$



No two edges cross each other at a non-vertex point

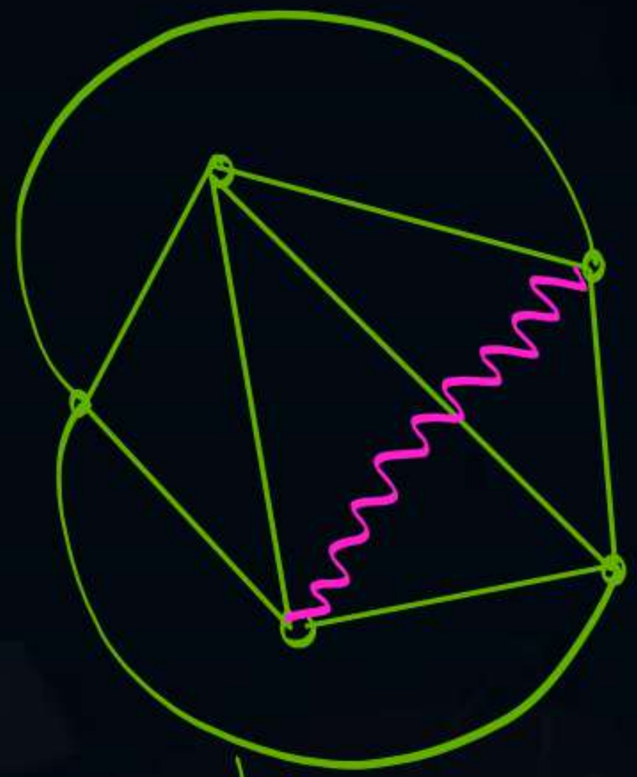
$\therefore$ K_4 is a planar graph

Topic : Planar graph

K_1 K_2 K_3 K_4

$\underbrace{\hspace{6em}}$

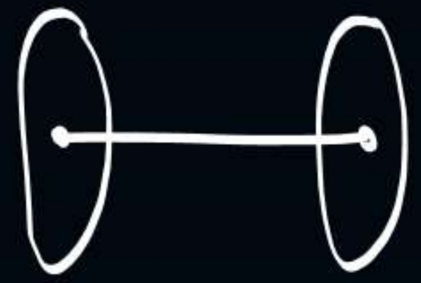
K_n is a planar graph if and only if $n \leq 4$



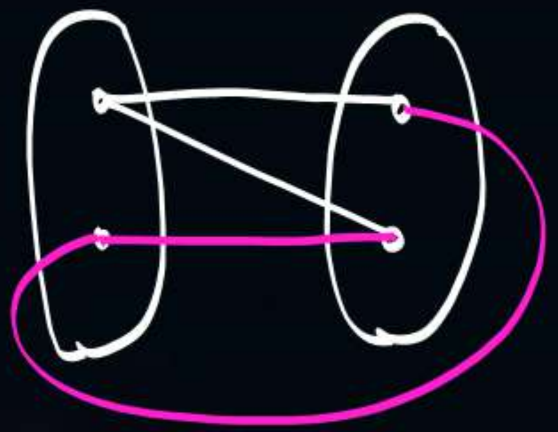
K_5

K_5 is a non-planar graph

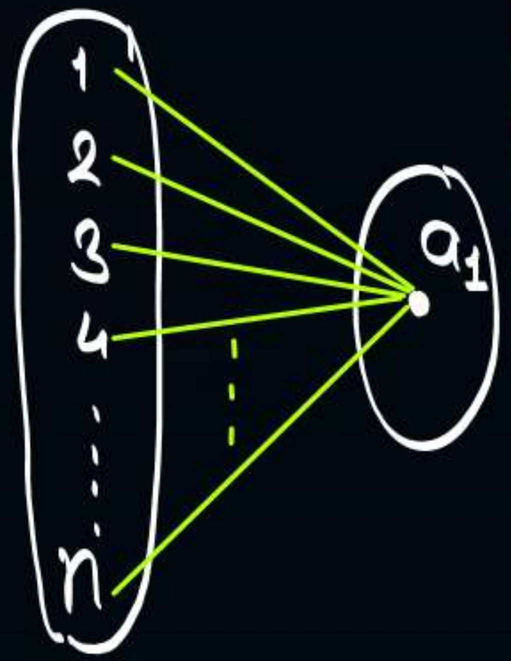
Topic : Planar graph



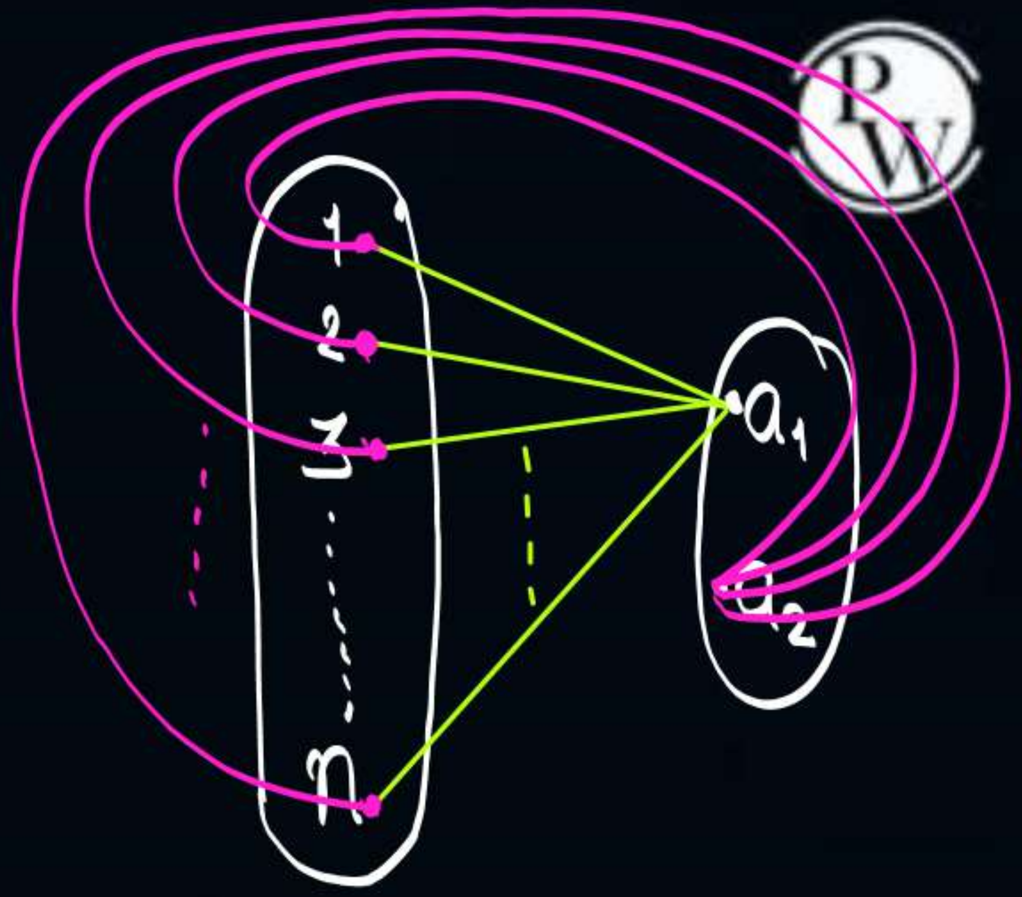
$K_{1,1}$ is planar



$K_{2,2}$ is planar

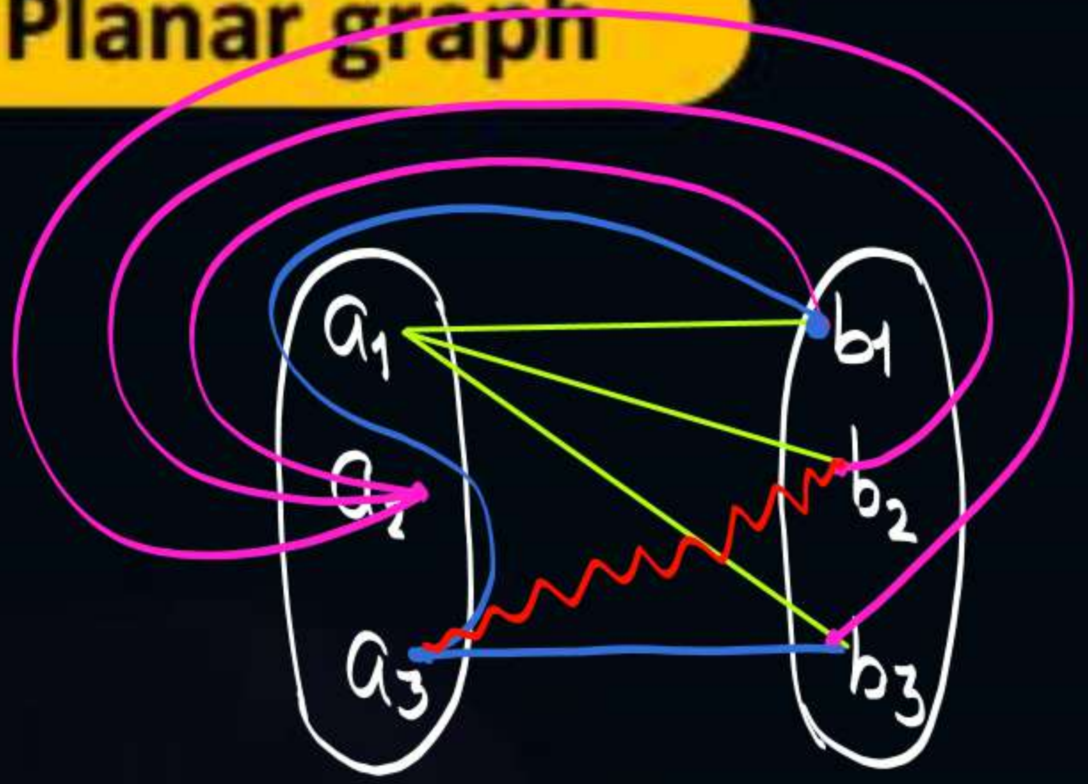


$K_{n,1}$ is planar, and similarly $K_{1,n}$ is also planar



$K_{n,2}$ is planar, and similarly $K_{2,n}$ is also planar

* Check whether $K_{3,3}$ is a planar graph or not?



$K_{3,3}$
is not a planar graph

* Complete bipartite graph.
$K_{m,n}$ is planar
if and only if
either $m \le 2$ or $n \le 2$

Note :-

* (1) K_5 is the non-planar graph with minimum number of vertices.

it has only 5 vertices but 10 edges

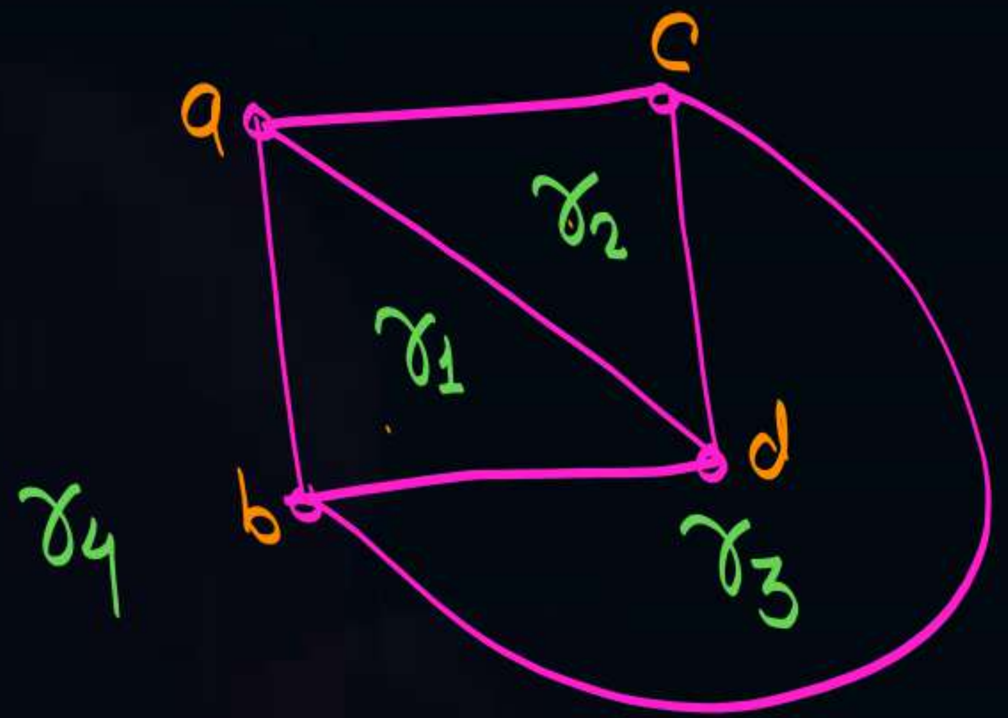
* (2) $K_{3,3}$ is a non-planar graph with minimum number of edges.

it has only 9 edges. but 6 vertices

Topic : Region / Face

- Every planar graph divides a plane into connected areas called as regions of the plane or faces of the planar graph.

eg.



- Here r_1, r_2 and r_3 are bounded regions (interior regions)

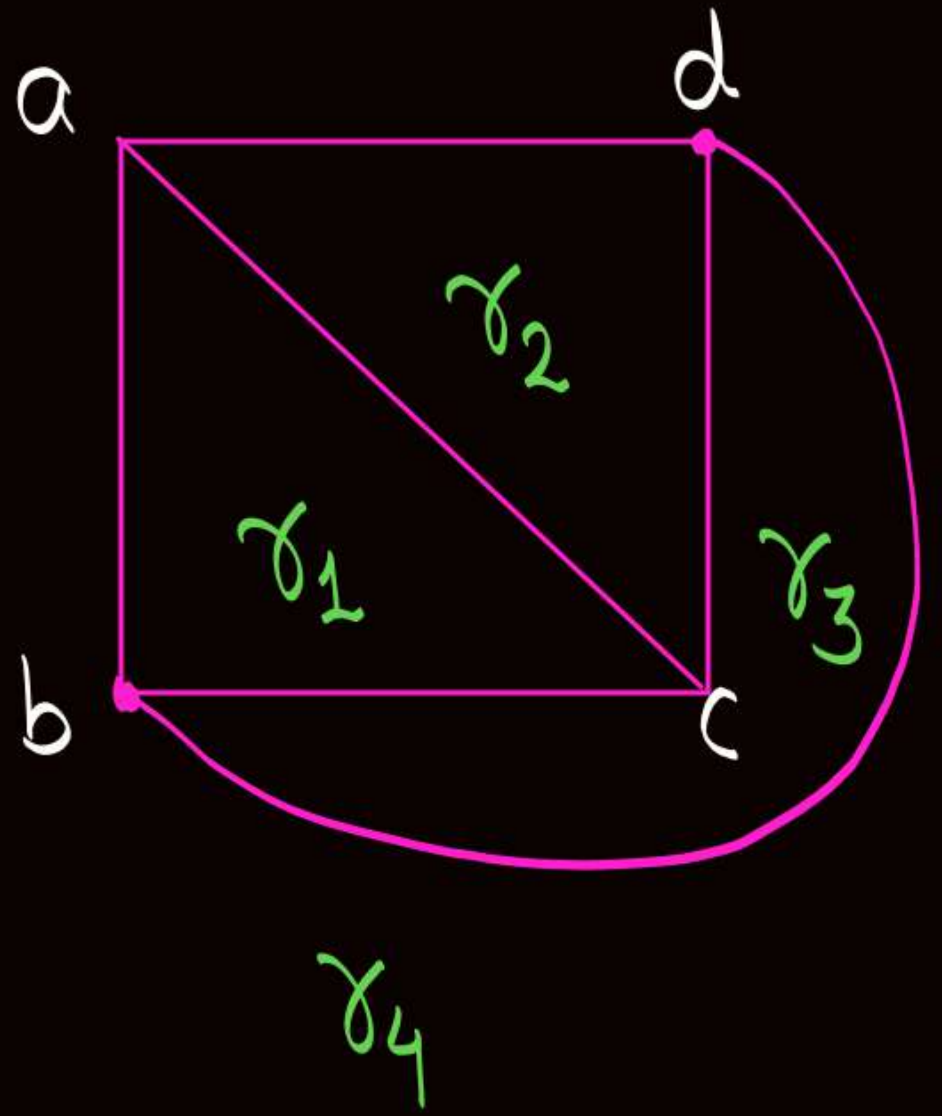
Whereas,

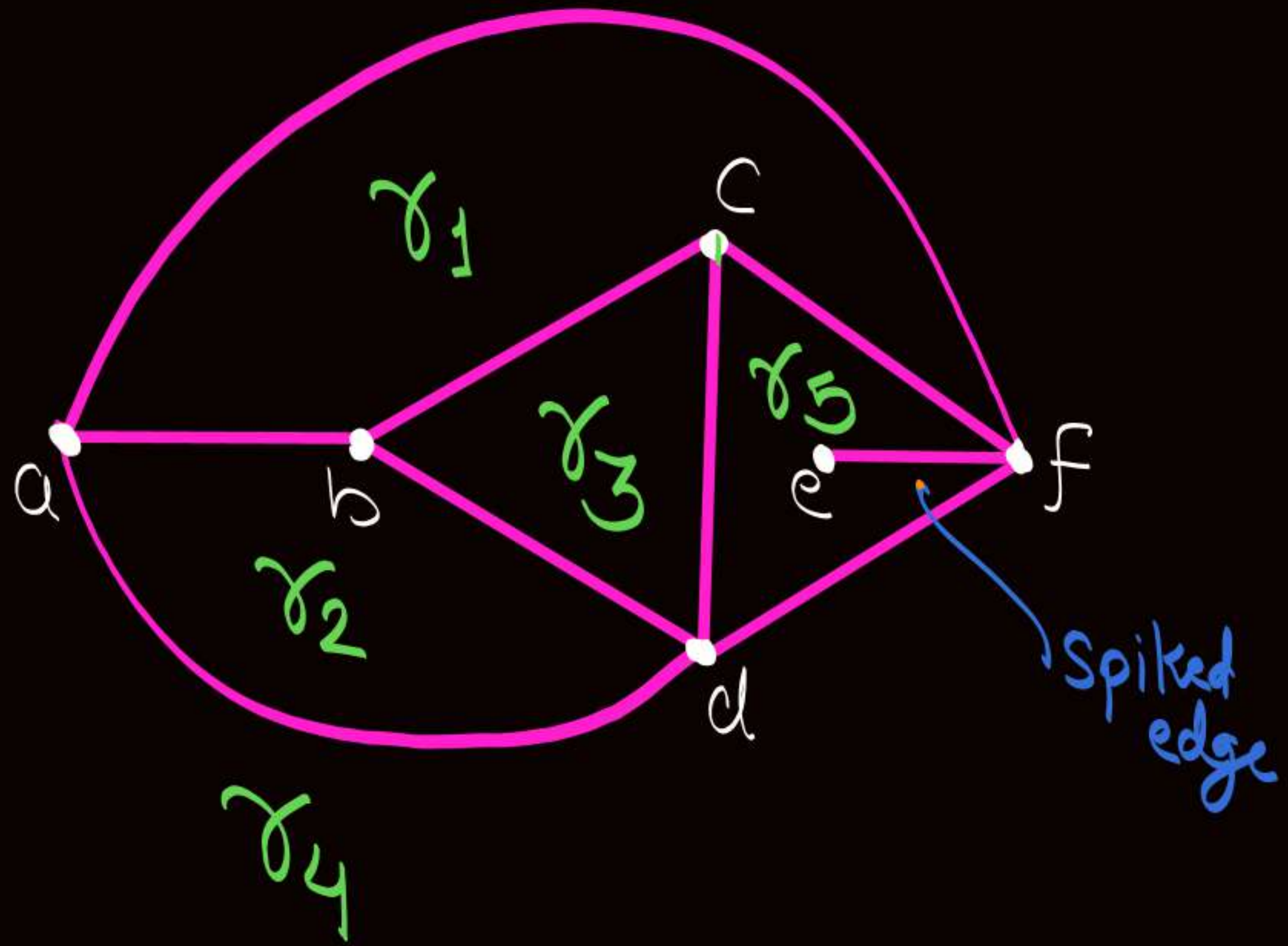
'r_4' is unbounded region or (Exterior region)

Topic : Region / Face

4. In any planar graph there will be exactly '1' Unbounded (Exterior) region, all other regions of the planar graph will be bounded (interior) regions.

+ $\boxed{\text{Number of bounded regions (faces) in a planar graph} = \text{Total no. of regions in that planar graph} - 1}$


a
d
b
c
r_1
r_2
r_3
r_4


a
b
c
d
e
f
r_1
r_2
r_3
r_4
r_5
spiked edge

Topic : Degree of a region/face

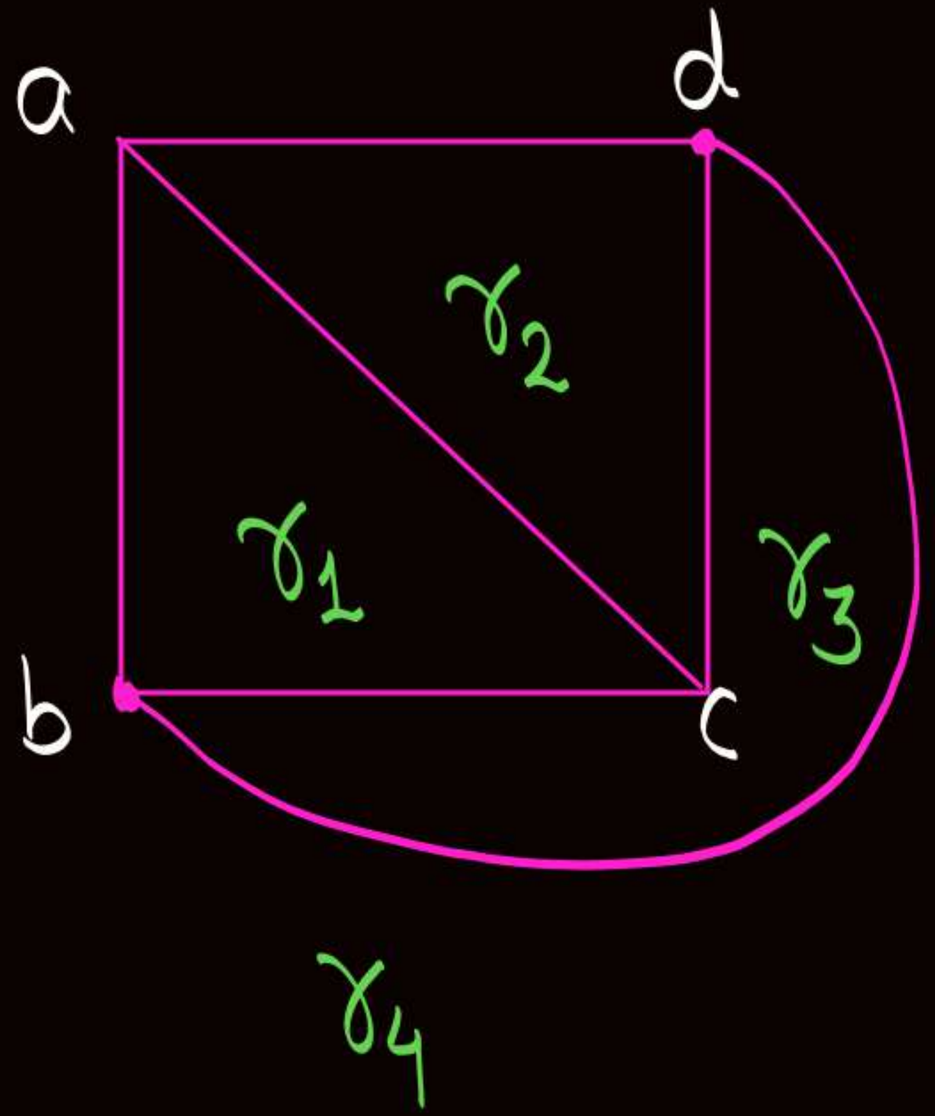
* Degree of a region "r" Can be denoted by $deg(r)$, and it is defined as.

$deg(r)$ = No. of edges Enclosing the region 'r'

(or)

No. of edges exposed to region "r"

eg:-



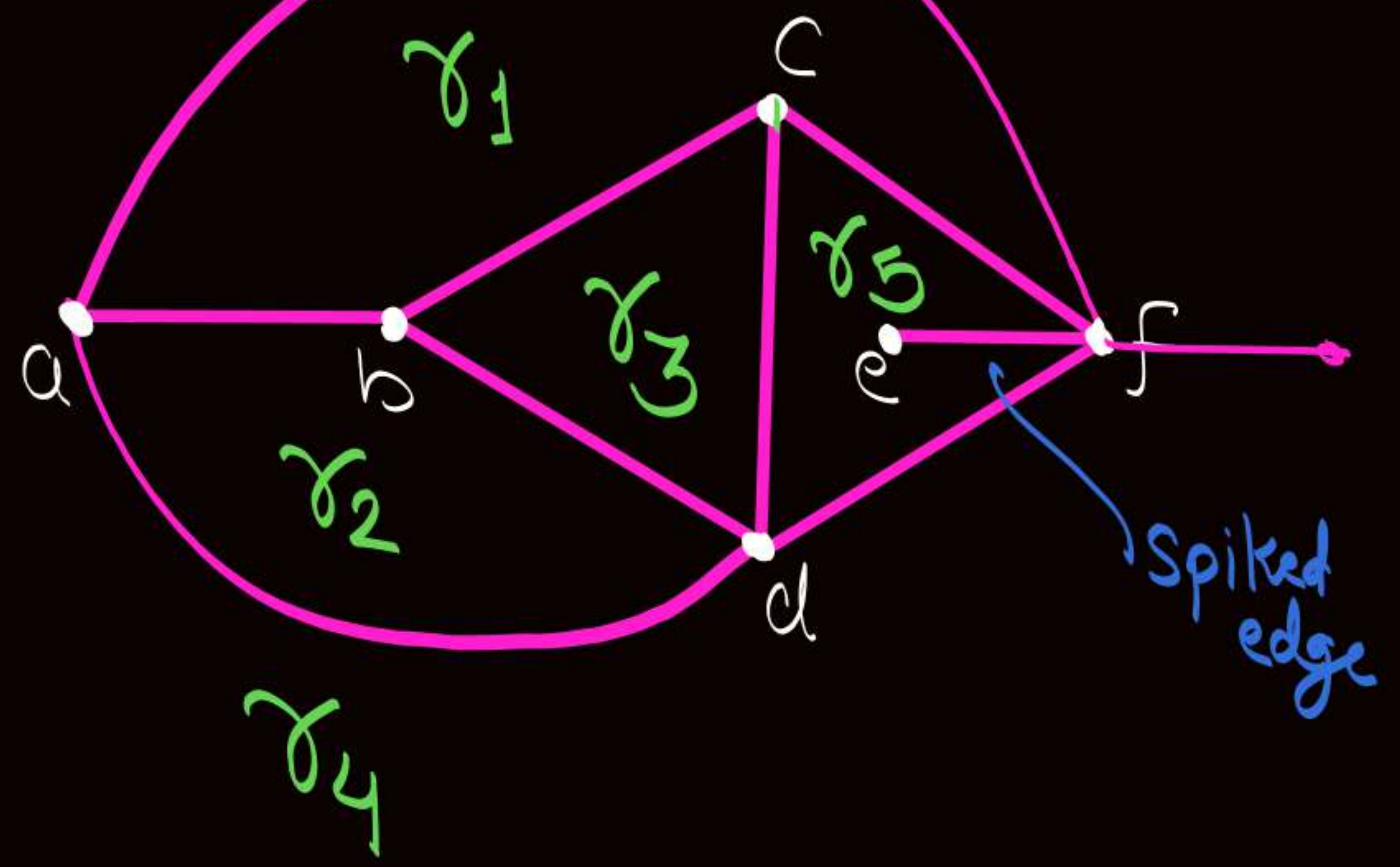
$\deg(r_1) = 3$

$\deg(r_2) = 3$

$\deg(r_3) = 3$

$\deg(r_4) = 3$

eg:



$\deg(r_1) = 4$

$\deg(r_2) = 3$

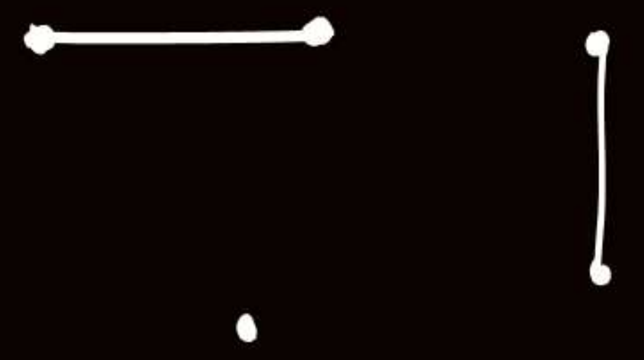
$\deg(r_3) = 3$

$\deg(r_4) = 3$

$\deg(r_5) = 3 + 2 = 5$

w.r.t. spiked edge

Note: Spiked edge will be counted as two edges to find the degree of that region eg:



Consider above Graph G

$\Downarrow$

it is a planar graph as no two edges cross each other at a non-vertex point.

But it is a planar graph with only '1' region; lets call it region 'r'

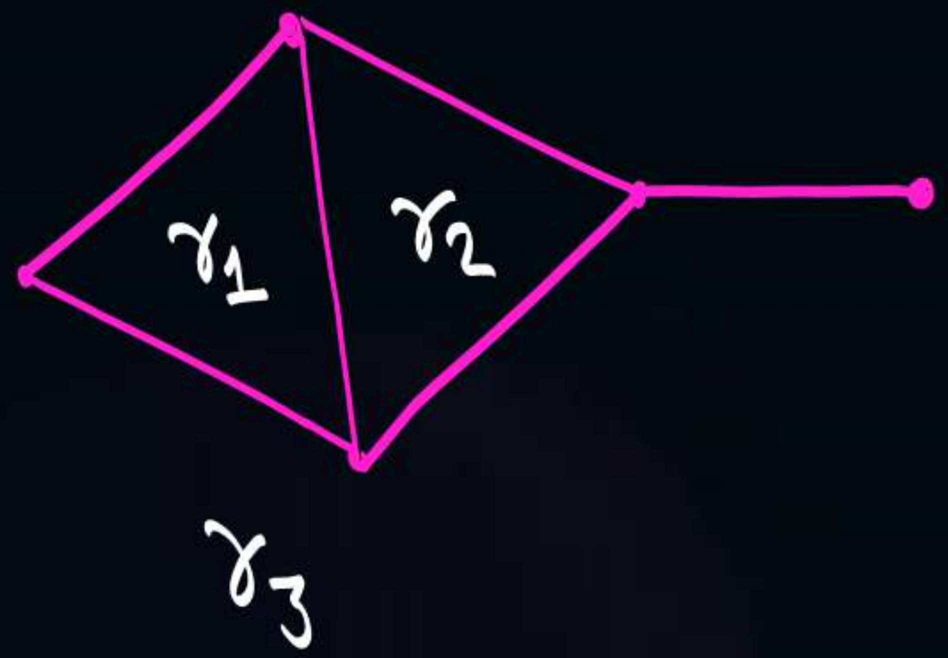
$$\deg(r) = 2 + 2 = 4$$

w.r.t. 1st spiked edge

w.r.t. 2nd spiked edge

Topic : Sum of degree theorem

W.r.t. degree of regions



Let $|R|$ denotes the number of regions in the planar graph

$$\therefore \sum_{i=1}^{|R|} \deg(r_i) = 2 * |E|$$

$\deg(r_1) + \deg(r_2) + \deg(r_3)$

$3 + 3 + 6 = 12$

$= 2 * 6$

$= 2 * |E|$

Topic : Sum of degree theorem

* Corollary 1:- No. of regions of odd degree are always even

* Corollary 2:- In a planar graph G, if degree of each region is Exactly 'K' then

$$\boxed{K \cdot |R| = 2|E|}$$

Topic : Sum of degree theorem

Corollary 3: In a planar graph G if degree of each region is at least 'K' {i.e. $\geq K$} then

$$\boxed{K|R| \leq 2|E|}$$

$$\deg(r_1) + \deg(r_2) + \cdots + \deg(r_{|R|}) = 2|E|$$

$$(K+x_1) + (K+x_2) + \cdots + (K+x_{|R|}) = 2|E|$$

$$\boxed{K|R| + \underbrace{\sum_{i=1}^{|R|} x_i}_{\geq 0} = 2|E|}$$

Topic : Sum of degree theorem

Corollary 4: In a planar graph G if degree of each region is at most 'K' {i.e $\leq k$} then

$$\boxed{K|R| \geq 2|E|}$$

$\deg(r_1) + \deg(r_2) + \cdots + \deg(r_{|R|}) = 2|E|$

$(K - x_1) + (K - x_2) + \cdots + (K - x_{|R|}) = 2|E|$

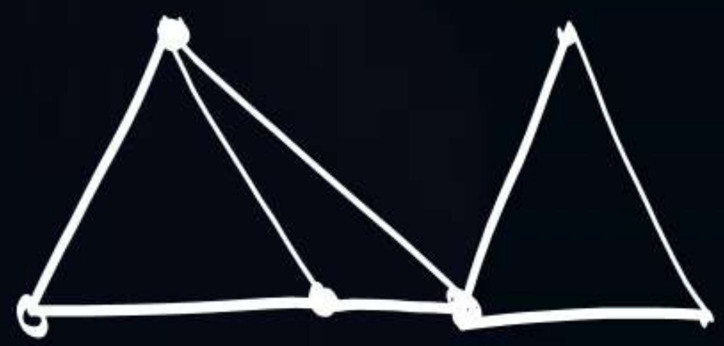
$K|R| - \sum_{i=1}^{|R|} x_i = 2|E|$

≥ 0

Topic : Simple planar graph

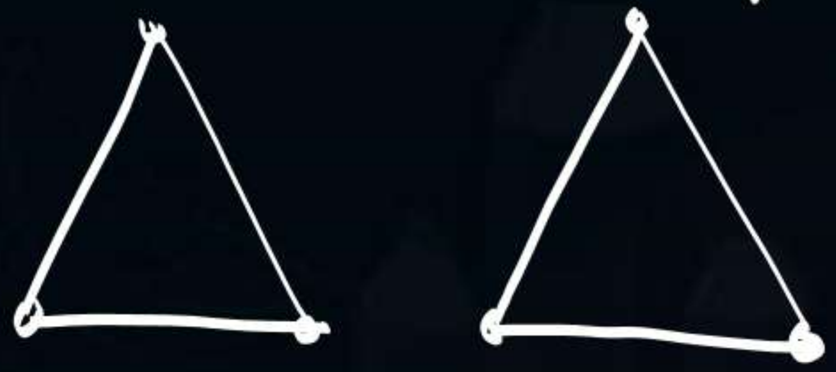
A planar graph with no loop and no parallel edges is called a simple planar graph {It may or may not be Connected.}

eg: G_1



Connected Simple Planar graph

eg: G_2



It is a Simple planar graph, but not a Connected graph

Topic : Simple planar graph

Note:- In a planar graph with two or more regions.
{i.e, in a planar graph with at least two regions}

deg (region) = 1, if and only if there is a loop.

eg:

$deg(r_1) = 1$
& $deg(r_2) = 1$

deg (region) = 2, if and only region is w.r.t. Parallel edges

eg:

$deg(r_1) = 2$
& $deg(r_2) = 2$

Topic : Simple planar graph

- In a simple planar graph with two or more regions degree of each region will be at least '3'

i.e. In a simple planar graph with two or more regions

$$\boxed{3|R| \leq 2|E|}$$

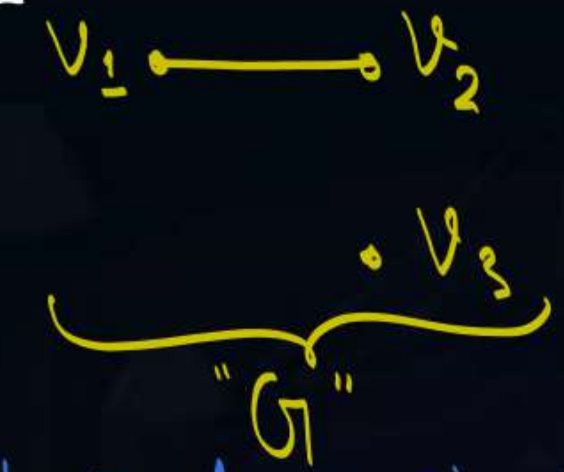
will always hold true

because deg. of each region is at least '3'

eg 1:-

It is a simple Planar graph with only '1' region and deg(region) = 0

eg 2:-



It is also a simple Planar graph with only '1' region and deg(region) = 2

Topic : Euler's equation (For Connected Planar graph)

- For any connected planar graph G, {may or may not be a simple graph} following Equation will hold true

$$\boxed{|V| + |R| = |E| + 2}$$

← and it is called Euler's Equation for connected planar graph

Where, $|V|$ = No. of vertices

$|R|$ = No. of regions

$|E|$ = No. of edges

$|V|+|R|=|E|+2$ Will not hold true if Planar graph is disconnected

Some author define the Euler's equation using the following format

$$\boxed{n+f-e=2}$$

Where
n = No. of vertices
f = No. of faces
e = No. of edges

Note 1:- In a simple planar graph {may or may not be Connected}

$$3|R| \leq 2|E| \text{ ——— eq}^n \text{ (1)}$$

will always hold true

Note 2: In a connected planar graph {may or may not be simple}

$$|V| + |R| = |E| + 2 \text{ ——— eq}^n \text{ (2)}$$

will always hold true

Topic : Simple connected planar graph

Simple Connected planar graph $\Rightarrow$ Simple planar graph + Connected Planar graph

$\downarrow$ $\left(\therefore 3|R| \leq 2|E| \text{ will hold true} \right)$ and $\downarrow$ $\left(\therefore |V| + |R| = |E| + 2 \text{ will also hold true} \right)$

1. In a simple planar only eqn 1 is guaranteed to hold true, but eqn 2 may or may not hold true

2. In a Connected Planar graph only eqn 2 is guaranteed to hold true, but eqn 1 may or may not hold true

3. If graph is a simple Connected Planar graph, then both eqn 1 & eqn 2 will hold true

Topic : Simple connected planar graph

Simple Connected planar graph $\Rightarrow$ Simple planar graph + Connected Planar graph

$\therefore\ 3|R| \le 2|E|$ — eqⁿ ①

$\therefore\ |V| + |R| = |E| + 2$ eqⁿ ②

from Eqⁿ ① $3|R| \le 2|E|$ $\therefore\ |R| \le \frac{2|E|}{3}$

Put this value of $|R|$ into eqⁿ ②

i.e. $|V| + \frac{2|E|}{3} \geq |E| + 2$

$|V| \geq \frac{|E|}{3} + 2 \Rightarrow \boxed{|E| \le 3|V| - 6}$ — eqⁿ ③

$e \le 3n - 6$

from eqⁿ ② $3|R| \le 2|E|$ $\therefore\ |E| \geq \frac{3|R|}{2}$

Put this value of $|E|$ in to eqⁿ ②

i.e. $|V| + |R| \geq \frac{3|R|}{2} + 2$

$|V| \geq \frac{|R|}{2} + 2 \Rightarrow \boxed{|R| \le 2|V| - 4}$

eqⁿ — ④

Topic : Note

In a simple Connected planar graph all four equations will always hold true

$$3|R| \le 2|E| \quad \text{——— } ①$$

$$|V| + |R| = |E| + 2 \quad \text{——— } ②$$

$$|E| \le 3|V| - 6 \quad \text{——— } ③$$

$$|R| \le 2|V| - 4 \quad \text{——— } ④$$

2 mins Summary

- Topic: Planar graphs
- Topic: Region and degree of regions
- Topic: Sum of degree theorem for regions
- Topic: Simple planar graph
- Topic: Euler's equation

Simple Connected Planar graph

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THANK - YOU

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